

Why Atoms Bond

Independent Application Answer Sheet - Notebook Evidence

Student name: _____ Date: _____

Directions

Use the lesson and worked examples. Complete every item. Submit this PDF with your U3 L1 Notebook Evidence Submission.

Part A - Predict the attraction

PhET test	Distance	Force value	Direction / evidence
$q_1 = +1e$; $q_2 = -1e$	40 pm		
$q_1 = +1e$; $q_2 = +1e$	40 pm		
$q_1 = +1e$; $q_2 = -1e$	20 pm		

Part B - Explain why bonding lowers energy

1. In your own words, explain why atoms can become more stable when a bond forms.

2. Compare an isolated sodium atom and chlorine atom with Na^+ and Cl^- together. Include valence electrons, charge, attraction, and stability.

3. Correction check: If your explanation says atoms bond only to 'get eight electrons,' revise it to include electrostatic attraction and lower potential energy.
